ANTARA THAPLIYAL

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PROFESSIONAL SUMMARY

Engineer who works directly with users to understand messy, real problems and ships end-to-end solutions: data integration and modeling, backend services, and AI agents. 2 years delivering production systems for an insurance client at TCS and a GenAI engineering internship at Otsuka Pharmaceutical; MS in Computer Science at Rutgers (4.0 GPA, graduating Dec 2026). Comfortable owning a problem from first conversation to deployment and explaining trade-offs to technical and non-technical stakeholders.

TECHNICAL SKILLS

Languages: Python (FastAPI, pydantic, pytest), SQL, JavaScript, HTML/CSS

Data Integration & Modeling: PostgreSQL, relational data modeling, data pipelines, schema mapping and normalization; semantic layers and documented knowledge bundles (tables, join keys, metric definitions) so systems query data correctly

AI & Agents: LLM agents and orchestration (LangGraph, LangChain, CrewAI), tool use, MCP servers, RAG (FAISS, Chroma), evals (LLM-as-judge, RAGAS), guardrails; Claude via AWS Bedrock

Integrations & Delivery: REST API integrations (auth, retries, rate limiting), batch jobs, CLIs and internal tools, testing and debugging in production, technical documentation; Git

Cloud & Delivery: AWS (Bedrock, Lambda, S3, CloudWatch, IAM, cross-account access), Docker, CI/CD (Azure DevOps pipelines, GitHub Actions), Streamlit dashboards

AI-Assisted Development: Claude Code (daily), Cursor, GitHub Copilot

PROFESSIONAL EXPERIENCE

Otsuka Pharmaceutical, Princeton, NJ — Jun 2026 – Aug 2026

GenAI Engineer Intern – Gen AI Platform Team, AI-Enablement Org

- Built tools used across the company's AI platform, owning each **end to end**: gathered requirements from teams, finalized scope with them, then built, tested, and deployed. Shipped a **Code Review Agent** (Claude on AWS Bedrock) that pulls code via Azure DevOps REST APIs, runs deterministic checks (Bandit, Ruff, custom AST compliance checker) in a sandbox, and has the LLM write risk-rated reports; validated on a golden dataset; deployed via CI/CD.
- Upgraded the platform's agent CLI from **one-shot to multi-turn** and built MCP-accessible tools so anyone in the company could use agents easily; diagnosed and fixed cross-account AWS invocation failures blocking deployment.
- Built a **model-lifecycle agent** that pulls end-of-life dates directly from the AWS Bedrock API, applies tiered CRITICAL/WARNING alerting, emails managers on a schedule, and surfaces risk on a Streamlit dashboard.

Tata Consultancy Services (TCS) — Jul 2021 – Jun 2023

Software Engineer – Client Delivery, Insurance Client

- **Delivered production systems for an insurance carrier** in a regulated environment: built new insurance products and batch letter-generation flows for policy activation and annual renewals, with error handling, retries, and data contracts across third-party integrations.
- Worked directly with client stakeholders to turn requirements into build plans and **automated data pipelines** in Python and SQL with monitoring; guided junior team members; drove a **29% lift in sales** and **30% reduction in manual effort**.

Rutgers University, New Brunswick, NJ — Sep 2025 – Present

Instructor, CS110 & CS170

- Teach Python, web technologies, and data analysis to 80+ students per semester, many from non-technical backgrounds.

PROJECTS

Open Knowledge Format (OKF) | *Python, PostgreSQL, Claude* | 2026

- Modeled a synthetic insurance **PostgreSQL database** into a semantic layer of documented knowledge bundles (entities, join keys, metric definitions, playbooks) so a Claude agent answers business questions with correct queries instead of guessing.

Abilify Clinical Q&A Agent – Agentic RAG System | *Python, LangGraph, MCP, LangFuse, RAGAS, Claude, Chroma* | 2026

- Multi-agent RAG system for a high-stakes domain; parent-child chunking lifted retrieval accuracy **40%**, Cohere reranking raised answer relevancy **30%**; evaluation agent (LLM-as-judge, RAGAS) with regression detection; hallucination and PII guardrails; LangFuse observability.

Shadow Board – Multi-Agent Decision System | *Python, CrewAI, MCP, FastAPI* | 2026

- **5-agent** decision simulation with MCP integration and per-agent context pruning (**4s TTFT**); structured outputs, guardrails, and evaluation checkpoints behind a documented FastAPI layer.

EDUCATION

Rutgers University, New Brunswick, NJ — Expected Dec 2026

Master of Science in Computer Science | GPA: 4.0

G.B. Pant University of Agriculture & Technology — Jun 2021

Bachelor of Technology in Information Technology

CERTIFICATIONS & ACTIVITIES

- Certifications: Supervised Machine Learning; Introduction to Model Context Protocol; Claude Code in Action; Building with the Claude API.
- Women Who Code Leadership Summit 2026, Emcee Team; Astitva NGO Volunteer Educator (2+ years).